

Chapter 6

ESD Controls in Cleanroom Environments: Relevance to Particle Deposition

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6.1 INTRODUCTION

Contaminant particles are produced inside a high-quality cleanroom by personnel, production equipment and processes, and materials transfer in and out of the cleanroom. Unfortunately, all of these particle sources are usually close to the product. If surfaces are also charged, electrostatic attraction (ESA) tends to attract and adhere particles to the surface that would otherwise remain airborne in the cleanroom laminar air flow. Sub-micrometer-sized particles cause defects in semiconductor production similarly to the visual defects from dust particles on a photographic negative. With the trend increasing toward smaller feature sizes in semiconductor devices, the size of the “killer” particles also decreases. Smaller particles are more easily attracted and more difficult to remove because

of the static charge on surfaces and because the adhesion force per unit area increases as the particle size decreases.

Static charge also causes other production problems since the uncontrolled transfer of static charge can damage product directly. This is an electrostatic discharge (ESD) event that typically occurs in 1–10 ns. One result of this rapid transfer of energy is local heating, sufficient to melt the silicon or metallization of the semiconductor device and cause device failure.

Electrostatic discharge events are typically in the frequency range of 100 MHz–2 GHz, which is the same frequency range as the microprocessors that control the operations of most semiconductor production equipment. ESD events generate electromagnetic interference (EMI) that can cause a variety of equipment malfunctions from simple stoppage to erratic robotic operations that destroy products. Since EMI propagates through both radiation and conduction, an ESD event in one piece of equipment may affect the operation of other nearby equipment, making the source of the problem difficult to locate.

This chapter focuses on the problems of static charge in cleanroom environments as related to particle contamination and the methods to control static charging.

6.2 ELECTROSTATIC CHARGE PROBLEMS IN CLEANROOMS

The relationship between electrostatic charge and contamination control in high-technology cleanrooms is a strong one, representing a significant source of particle deposition on charged objects within the cleanroom. The rate of ESA of small particles to charged objects has been calculated for a controlled environment¹ and particle deposition data recorded in cleanroom environments have been found to be consistent with the calculation.^{2,3}

Several studies involved employing a high-voltage (HV) power supply to bias one wafer to a voltage of at least 2 kV while another was electrically connected to ground. In one of the studies,³ 200 mm wafers were exposed to the environment in an isolated part of an ISO Class 3 cleanroom. Human traffic through the area was eliminated by cordoning off the area where the exposure was done. In order to assure that a statistically valid sample was acquired, an exposure time of 6 weeks was employed. The wafers were scanned using a Tencor Surfscan with a particle size threshold of 0.2 μm . The Surfscan results are shown in Figure 6.1. While the numbers of particles observed on the two wafers are not representative of real world contamination levels for semiconductor processing, the ratio of the number of particles on the neutral wafer (3,389) compared to the number of particles on the wafer at a voltage of 2000 V (22,764) is accurate. In this case, the ratio is 6.7:1.

The calculation of electrostatically driven contamination sums the components of the deposition velocity due to each deposition mechanism. Deposition velocity, loosely defined, is the rate at which particles move toward the surface

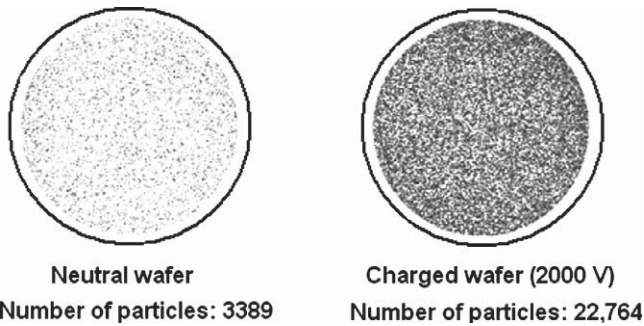


FIGURE 6.1 Comparison of particle contamination on a charged (a) and a neutral (b) silicon wafer.

of the object due to the forces on the particle. Particles deposit on wafers, flat panel displays, disk drive components, and equipment surfaces due to gravity, diffusion, and other forces. High efficiency particulate air (HEPA) filtration attempts to keep particles out of the cleanroom, and laminar airflow is designed to minimize the sedimentation rate of contaminating particles by entraining them in the air flow. Even though a cleanroom is maintained at a high level of cleanliness, particles are still generated within the cleanroom due to personnel, equipment motion, and processes.

The forces on the particle are aerodynamic (viscous drag), gravitational, diffusion, and electrostatic in nature. The gravitational forces decrease as the size (mass) of the particle decreases. Similarly, the magnitude of the viscous drag on the particle also is smaller for small particle sizes. In contrast, the magnitude of the diffusional forces is greater for smaller particles because the greater efficiency of momentum transfer to particles that are closer in mass to the gas atoms that strike them.

Calculation¹ shows that the force of ESA exceeds the other physical forces for the conditions normally present in a cleanroom. The conditions include surface voltages of several thousands of Volts and particles in the micrometer to sub-micrometer range. The calculation assumed that the particles in the air were neutral on the average but had a distribution of charges with a width which was monotonically related to the total number of electrons on the particle. Thus, larger particles have the possibility of more charge on them and can experience greater electrostatic forces but also are more massive. Thus, the deposition velocity due to electrostatic forces exhibits a maximum for a specific size.

Theoretically, increased particle deposition due to electrostatic forces will occur at any voltage, but at field strengths of 200 V/cm or more, the effects are significant and easy to measure. Field strengths of 4000 V/cm or more are not uncommon in cleanrooms.

Figure 6.2 illustrates some of the forces that act to attract particles to a surface.⁴ For small particles (0.01–1.0 μm in size), electrostatic forces are the primary cause of increased particle deposition.



What size of particle is to be expected in a modern high-technology clean-room? [Figure 6.3](#) gives some insight into the size range to expect. The figure shows that the nature of a HEPA filter is to stop virtually all particles that are incident upon it, but that the efficiency of the filter is lowest in the near sub-micrometer range ($\sim 0.1\text{--}0.2\text{ }\mu\text{m}$). In this size range and with the forces

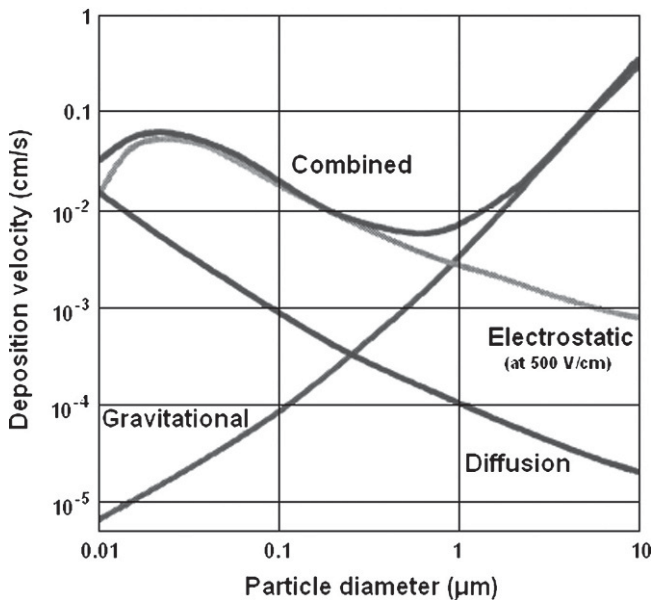


FIGURE 6.2 Deposition velocity as a function of particle diameter for the principal forces acting on a particle.

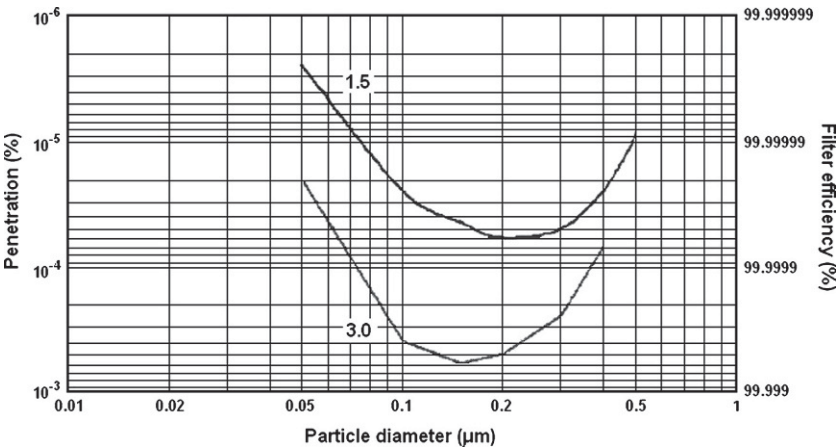


FIGURE 6.3 Filter efficiency as a function of particle size for two different filter configurations. (Courtesy of Camfil-Farr Corporation, Riverdale, NJ, USA.)



shown in Figure 6.2, it is expected that the major contributor to contamination at 500 V/cm is ESA.

How significant are the calculations and the measurements discussed above? Is ESA a significant factor in high-technology manufacturing? In a typical cleanroom used in high-technology manufacturing which has no electrostatic charge control program, insulators in a room (e.g., reticles, wafers with oxide coating, and disk drive media) typically achieve voltage levels of 5-20 kV. Thus, based upon the data presented above, the majority of the contamination is electrostatically driven. Often this major factor goes unnoticed because the cleanliness levels are so good in the cleanroom. In sub-ISO Class 3 cleanrooms, the number of “particle adders” (particles-per-wafer-pass or PWP) in a typical modern process can be as low as 0.1 μm (>75 nm). Over the course of processing many layers of the product this can be 2-20 particles, still a very small number. If the average static charge level in the process is only 500 V owing to the fact that there is static control in some parts of the process delivered by the process tool manufacturer, the part of the contamination contributed by ESA would be quite low (~ 0.25 -10 particles). In a process with such low-contamination levels, it is very difficult to devise a method for measuring the contribution due to one particular source. Nonetheless, the economic impact of removing this contamination source is profound. For example, we assume only one of the static-attracted particles causes a die loss on 10 semiconductor wafers. If there were 100 dies on the wafer this would represent a yield loss of 0.1%. If the wafer fab produced 500,000 wafers per year, this would be a loss of 50,000 dies per year. With good dies selling for up to US\$500 each, this is a significant loss. In reality, static-attracted particles cause greater losses than that used in this example.

Electrostatic attraction and its effect on contamination control is not the only negative effect of static charge. As mentioned earlier, static charge is responsible for ESD due to the discharge from one object to another. Such a discharge can cause physical damage to a product which experiences such a discharge owing to either electrical overstress or due to damage caused by the energy deposit of the discharge (spark). ESD is particularly important in the case of disk drive assembly (manual handling of magnetoresistive (MR) heads) and in photolithography (damage to the reticle caused either by charge on the reticle, or by induced discharges caused by electric fields from other charged objects in the proximate environment). Most recently, semiconductor wafers with the smallest feature sizes (≤ 0.15 μm) have exhibited gate oxide punch-through due to processing steps that are known to cause high charging such as cleaning and certain wet processing steps.

The remaining problem caused by ESD is robotic malfunction. When a metal-to-metal discharge takes place, the frequency content of the discharge is extremely high ($\sim$ multiple GHz). The discharge dissipates much of its energy in the form of EMI transients which are within the bandpass of microprocessors controlling the robots. Occasionally ($\sim$ daily to weekly), such a discharge can

cause the microprocessor to execute meaningless or invalid instructions. When this happens, the robot will either stop and display a difficult-to-interpret error message or, worse, exhibit some strange behavior like bumping into a wall or trying to insert a wafer into an incorrect location. Most often this behavior is assumed to be a software bug. While this may be the case, the problem can also be the result of EMI transients generated by ESD.

Electrostatic charge control in cleanrooms must prevent product damage, equipment malfunctions, and ESA caused by charge accumulation.

6.3 STATIC CHARGE GENERATION

Whenever two dissimilar surfaces placed in close contact are separated, one surface loses electrons and becomes positively charged, while the other surface gains electrons and becomes negatively charged. This is known as *triboelectric charging*. Any material (solid or liquid) may be charged triboelectrically by friction, separation of materials, or fluid flow. The magnitude of the charge will be affected by the surface condition, area of contact, speed of separation or rubbing, and the humidity. Whether the material remains charged depends on its conductivity and the availability of a path for the charge to flow to ground. If charge is allowed to accumulate on a material, it may attract and bond particles to its surface.

The principle of triboelectric charging is related to the work function of each material. This is defined as the energy required to remove an electron from the surface of the material. Since the work function is related to the electronic energy levels of the material, the work function of each material is unique. Thus, any time two materials are placed in contact with each other, electrons are transferred from one material to the other. The amount of transfer is dependent upon the nature of the surfaces and the nature of the contact. Smooth surfaces with a significant amount of contact pressure and relative motion (sliding) result in a greater amount of charge transfer.

When the materials are separated, the capacitance between them decreases and the potential difference (voltage) between them increases. Thus, the mechanism of triboelectric charging results from contact of dissimilar materials and separation of the materials afterward.

The environment of a cleanroom lends itself to increased levels of tribocharging and decreased dissipation of the charge from the surface of insulators as compared with charge levels on insulators in a conventional room. For example, any contamination that exists on the surface of an insulator provides a discharge path to ground for the charge. In a cleanroom, all objects are wiped down to eliminate this discharge mechanism.

Studies⁵ have also shown the relationship between humidity and tribocharging. In any location, the humidity will achieve an equilibrium between water vapor in the air and on the surface of each material. At lower humidity [$\sim 35\text{--}45\%$ relative humidity (RH)], the equilibrium condition dictates that less water vapor

will be present on the surface of the material. This water vapor modulates the work function of the material, shifting the value between the work function of the material and the work function of water. Thus, materials in a low humidity environment exhibit more triboelectric charging than those that are in a higher humidity environment. Since cleanrooms are maintained at low humidity, more charging is observed in the cleanroom than in a conventional room.

Static charge is also generated by *induction*. Static charge on an object can create or “induce” opposite polarity charges on the surface of another object by causing the positive and negative charges on the object to separate. As with charge generated triboelectrically, the induced charges can attract particles and contact with ground can result in damaging ESD events and EMI.³ An example of contamination resulting from induction is when a wafer is stored in a highly charged container such as a front opening unified pod (FOUP) assumed to be negatively charged for discussion purposes. If wafers without an oxide coating are held in an inner support structure which is conducting (as is the case with a FOUP) and connected to ground, the charge on the wafer support structure will separate and the negative charge will be displaced from the wafers to the support structure and from there to ground. When the container is removed from the ground structure, the wafers will be charged positively and present a contamination issue.

6.4 INSULATORS VERSUS CONDUCTORS

All materials can be characterized as either electrical conductors or as electrical insulators. A conductor is a material through which an electrical current can pass, while an insulator is a material through which an electrical current cannot pass. There is a good deal of variation in the electrical properties of conductors (electrical resistivity) that allows electrical conductors to be further classified as conductors (good conductors) and as dissipative (poor conductors), but both are differentiated from insulators by their ability to sustain an electrical current. Electrical resistivity is a measure of the voltage that must be applied to cause a unit current to flow through a cubic sample of a material 1 m on a side. Surface resistivity is a similar parameter corresponding to current passing along the surface of a material rather than through a bulk sample. It is defined in terms of the voltage required to pass a unit current through a square sample of the material. Bulk resistivity is measured in units of $\Omega\cdot\text{m}$ or $\Omega\cdot\text{cm}$ and the surface resistivity (relating to currents passing along the surface of a sample) is measure in Ω/square . The range of values of resistivities for different classes of materials is shown in Table 6.1.

- *Conductive materials*⁶ are defined as those having a surface resistivity $<1 \times 10^5 \Omega/\text{sq}$ or a volume resistivity $<1 \times 10^4 \Omega\cdot\text{cm}$. With a low electrical resistance, electrons flow easily across the surface or through the bulk of these materials. Charges go to ground or to another conductive object that

TABLE 6.1 The Resistivity of Different Classes of Materials		
Material Type	Surface Resistivity (Ω/sq)	Volume Resistivity (Ω-cm)
Conductive	$<1 \times 10^5$	$<1 \times 10^4$
Dissipative	$\geq 1 \times 10^5, <1 \times 10^{12}$	$\geq 1 \times 10^4, <1 \times 10^{11}$
Electrostatic shielding	$<1 \times 10^4$	$<1 \times 10^3$
Insulative	$\geq 1 \times 10^{12}$	$\geq 1 \times 10^{11}$

the material contacts. Metals like copper and aluminum have exceptionally low resistivities on the order of 10^{-8} Ω-cm, making them extremely good conductors of electricity.

- *Dissipative materials*⁷ have a surface resistivity equal to or $>1 \times 10^5$ Ω/sq but $<1 \times 10^{12}$ Ω/sq, or a volume resistivity equal to or $>1 \times 10^4$ Ω-cm but $<1 \times 10^{11}$ Ω-cm. For these materials, the charges flow to ground more slowly and in a somewhat more controlled manner than with conductive materials. Dissipative materials are used for ESD control because they discharge so slowly that they often do not cause ESD damage to microstructures that are manufactured in high technology cleanrooms.
- *Insulative materials*⁸ are defined as those materials with a surface resistivity $>1 \times 10^{12}$ Ω/sq or a volume resistivity $>1 \times 10^{11}$ Ω-cm.

Because of the incredible variation ($>10^{15}$) in the resistivity of materials, it is clear that the ability of an insulator to dissipate surface charge is so poor that it can be ignored. For a typical 30×30 cm² object located ~ 1 cm from a ground plane, the capacitance of the object is $\sim 10^{-10}$ F, so that the characteristic time to discharge will be given by the product of its resistance and its capacitance.⁹ For a conductor, this corresponds to a time of $\ll 1$ s. For a fairly good insulator ($>10^{13}$ Ω/sq), this corresponds to $>1,000$ s. For extremely good insulators like quartz, Teflon or SiO₂, the material can be assumed to NEVER discharge. Thus, in a cleanroom where high levels of charge are developed on such surfaces, they will build up to extremely high levels over a period of days. Grounding one of these extremely good insulators accomplishes nothing and the charge remains on the surface. The small amount of humidity present in the cleanroom would only shorten the discharge time to hundreds of hours. Grounding of conducting or dissipative materials is an important first step in cleanroom static charge control, but it is of no help in dealing with insulators. Details of how to ground the materials, utilization of dissipative materials, selection of insulators that have similar work functions to the materials that they contact (called tribo-matching),

and techniques for dealing with static charge developed on insulators is the discipline called electrostatic management.

6.5 CLEANROOM ELECTROSTATIC MANAGEMENT

A variety of methods have been developed to deal with static charge. The basic methods are grounding of all facility components (walls, floors, work benches, and equipment), appropriate use of conductive and static dissipative materials, and local or room ionization to control static charge on insulators. In addition to these basic methods, education of cleanroom personnel in the practice of static control, as well as auditing for compliance with the static control program, are also essential to the success of the program.

Modern cleanroom environments make extensive use of grounding with conductive and static dissipative materials to control electrostatic charge. Grounding prevents the storage of static charge on materials that are connected to ground. If a conductive or static dissipative material does become charged, connecting it to ground will remove this stored charge. To be effective in controlling static charge, conductive and static dissipative materials must be provided with a reliable path for the static charge to flow to ground. The success of any grounding method depends on the integrity of the ground path. In critical applications, ground path monitoring and periodic verification may be important aspects of a static control program.

By creating the ground path, the static charge on equipment, materials and personnel, can be rapidly and harmlessly neutralized. Ground connections should exist for most accessible surfaces of production equipment. It is particularly important to assure grounding of these surfaces within 30 cm of the static sensitive product. Care should also be taken with moving parts of equipment. A ground connection that exists when a robot handler is stationary may be lost when the robot is in motion. Flexible grounding connections are advisable rather than relying on conductive lubricants.

Static dissipative materials are used in the construction of cleanrooms and mini-environments to reduce the accumulation of static charge. Walls and floors of a cleanroom and panels of a mini-environment may be constructed from these materials. Static dissipative materials should be selected over conductive materials in order to slow down the charge removal process and prevent a damaging ESD event. They may be used for parts of equipment that come into contact with a sensitive product. Tweezers used in disk drive assembly and dissipative work surfaces used for reticle inspection stations are two examples of locations where dissipative materials are the best material choice. In contrast to metal tweezers, metal work surfaces, or insulative mini-environment walls, the cost of the dissipative material is higher, but often the payback in terms of decreased amount of ESD damage makes the dissipative materials a good investment. Static dissipative packaging materials are also used to shield the

product from static charge buildup or ESD damage as it is transported between process steps in the cleanroom.

When an object takes on a given amount of charge and then discharges to a sensitive product, the discharge can occur at different rates depending upon the resistivity of the material. A metal-to-metal discharge typically occurs in a sub-nanosecond interval, whereas a discharge of a dissipative material to a metal object or a dissipative-to-dissipative discharge can take multiple microseconds to occur. The slower discharge necessarily will have a lower peak current than the faster discharge of the same amount of charge. Due to the small size of modern electronics (sub-micrometer MR heads or deep sub-micrometer semiconductor feature sizes), the time required to dissipate heat from the structure is extremely short,¹⁰ on the order of nanoseconds. Thus, a discharge time from a dissipative object can be orders of magnitude longer and not cause damage, whereas the same discharge from a conductor is often lethal to the product. While it is important to employ static dissipative materials to minimize the peak current in a discharge, use of dissipative materials does not assure that the product itself will not charge.

It will also be important to verify that the materials are cleanroom compatible and do not present a contamination hazard. Particularly with construction materials used in cleanrooms, static dissipative properties may be sacrificed to achieve other features like dimensional stability, low outgassing, or low cost.

Grounding methods and static dissipative materials are also used to assure that charge does not accumulate or transfer from personnel to sensitive products or equipment. A variety of personnel grounding methods are employed using garments, booties, gloves, static dissipative flooring, and wrist or heel straps. For successful static charge control, all of these methods must be monitored and verified periodically.

Unfortunately, cleanrooms must also use many materials that are insulators, such as Teflon, various plastics, and quartz. Very often the insulating materials are an essential part of the product itself. It is not possible to remove the electrostatic charge on insulators by connecting them to ground. Most insulators are easily charged, retain their charge for long periods of time, and are often close to or part of the product. Teflon and quartz wafer carriers, oxide coated wafers, epoxy IC packages, insulated component leads, and glass hard disk media are examples of insulators present in the cleanroom. The requirements of cleanrooms preclude the use of carbon particles or chemical additives to insulating materials to make them static dissipative. Antistatic sprays and solutions can create a contamination problem. Humidity control was proposed in the past as a static control method, but it has been shown to be expensive and ineffective.

Neutralizing static charge on insulators and isolated conductors has become a necessity for achieving high product yields in cleanrooms. With few other methods available, this will often require the use of some type of air ionization. Using only the highly filtered cleanroom air, ionizers create clouds of both

positive and negative air ions to neutralize static charges wherever they may exist in the cleanroom environment.

6.6 AIR IONIZATION FOR STATIC CHARGE CONTROL

Air is a rather good insulator so any static charge that develops on a clean surface of a good insulator will remain on that surface indefinitely. Air ionizers are used to decrease the electrical resistivity of the air through the addition of ions which act as charge carriers. Air consists primarily of nitrogen, oxygen, carbon dioxide, and other trace gases. Air ions are gas molecules in air that have either lost or gained an electron. Air ionizers are used to introduce ionization into the atmosphere and have an effect similar to that of a lightning storm on the environment. Ions occur naturally in the air, but in a cleanroom the HEPA and ULPA (ultra low penetration air) filters serve to eliminate the ions from the air.

Positive and negative air ions are present in normal outside air, resulting primarily from radioactive decay of materials in the soil (such as uranium) and gases in the air (such as radon). But most air ions are stripped out of cleanroom air by HEPA filtration. This makes the air in the cleanroom very insulative and encourages the generation of static charge. Air ionizers restore and increase the level of air ions in the filtered cleanroom air. When the ionized air comes in contact with a charged insulating surface, the charged surface attracts air ions of the opposite polarity. As a result, the static charge on the insulator is neutralized. Air ions of both polarities are required for neutralization because both polarities of static charge are created in the cleanroom. This is shown in [Figure 6.4](#).

There are three methods that are commonly used to make air ions for charge control. These methods are corona, photoelectric, and radioisotope ionizations. While ions created by these methods are identical, generally each method has its

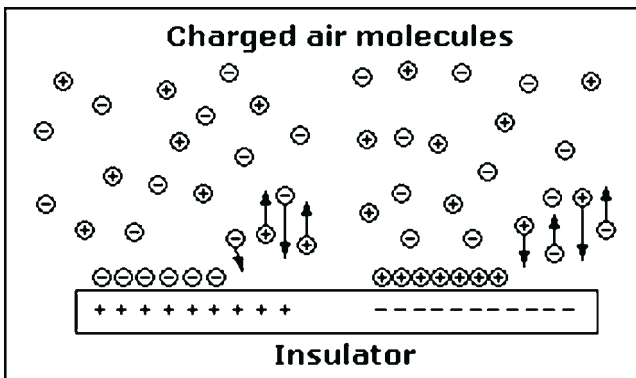


FIGURE 6.4 Neutralizing surface charges with bipolar air ionization.

own application. The tradeoffs involve issues of cleanliness level, ion distribution requirements, shielding, voltage balance (precision) level, and government regulations. Each type will be discussed below.

6.6.1 Corona Ionization

The most common method used to create air ions in cleanrooms is *corona ionization*. In corona ionization, a very high electric field is created by applying HV to one or more sharp emitter points. This technique involves the use of HV ($\sim 5\text{--}20\text{ kV}$). The voltage is applied to a set of sharp points and an intense electric field is established very near ($\sim 100\text{ }\mu\text{m}$) the points. This field accelerates free electrons to sufficiently high energy to allow them to ionize molecules that they collide with. When the voltage on the point is positive, positive ions are repelled into the environment; when the point is negative, negative ions are delivered to the environment. The basic principle of corona ionization for positive and negative polarities is shown in Figure 6.5. Corona ionizers can be made using AC or DC voltages. Each has an advantage. Both will be described below.

6.6.1.1 AC Ionizer

An AC ionizer uses a step up transformer to create the HV bias ($\sim \pm 7\text{ kV}$) required to make ions (Figure 6.6). The transformer creates both positive and negative voltage swings, so AC ionizers do not require any form of adjustment to work. The simplicity of an AC ionizer makes it the lowest cost method in most ionizer applications. Since an ionizer produces positive and negative ions in sequence from the same emitter point(s), these ions are separated in time by half of the period of the AC power line (i.e., $1/100$ or $1/120\text{ s}$). This means that the waves of positive and negative ions are rather close to each other, making loss through recombination a large factor. AC ionizers typically utilize fast

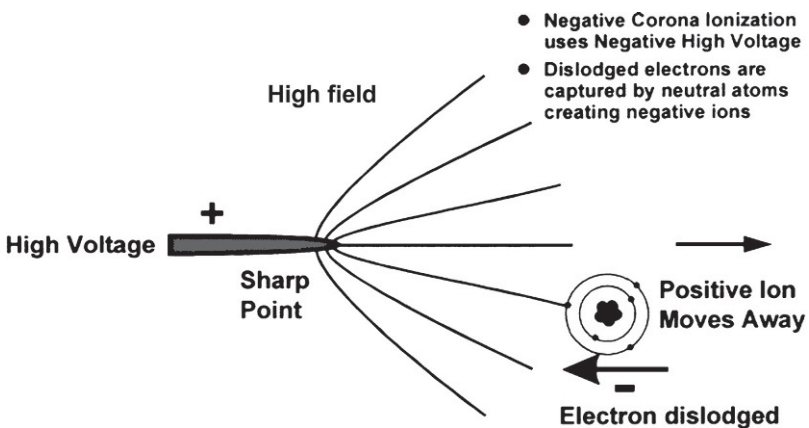


FIGURE 6.5 Principle of corona ionization.

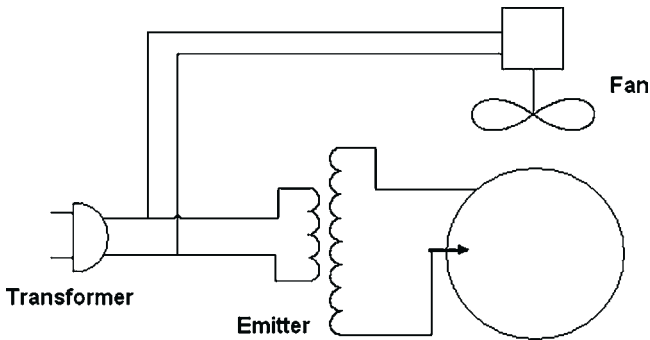


FIGURE 6.6 Schematic diagram of an AC-type ionizer.

airflow velocity to minimize recombination. This is not always desirable in a cleanroom environment. Increased emitter point current to offset recombination also increases contamination and limits an AC ionizer to ISO Class 5 cleanrooms or above.

Since the transformer secondary is well isolated from ground, the current drawn from the emitter point(s) during the positive and negative voltage excursions of the emitter (corresponding to positive and negative excursions of the AC power line) should be equal. Experience shows that the actual offset voltage as measured with a Charged Plate Monitor (CPM) is typically <10 V.¹¹ Unfortunately, it has also been shown, that a CPM is not capable of accurately recording voltage excursions faster than 1 Hz.¹² Although the instrument has the bandwidth to track such signals, its plate cannot slew fast enough to follow signals that vary at 50–60 Hz. In contrast, the small size of objects like an MR head make them capable of tracking the voltage excursions driven by the time varying electric fields from an AC ionizer. The actual voltage excursions on these objects may be significantly greater than the balance voltage measured by the CPM. Thus, for small objects with extreme voltage sensitivity, such as an MR head, AC ionizers are not recommended.

6.6.1.2 DC Ionizer

A DC ionizer employs a HV power supply to create the necessary voltage. DC ionizers use separate emitter points connected the positive and negative DC HV power supplies to generate the ions. In order to provide equal numbers of positive and negative ions from separate sources, DC ionizers need some form of control to maintain this balance. For the demanding requirements of electrostatic management of production lines where GMR (giant MR) heads are handled, the control must be active feedback to account for variations in the environment and for any wear that occurs over the life of the ionizer. A schematic diagram of a DC type ionizer is shown in Figure 6.7. Because of the greater sophistication and control systems needed for DC ionizers, the DC

ionizing systems are more expensive to manufacture. Owing to the fact that the positive and negative emitters are well separated from each other, recombination is a lesser effect and the DC ionizer can sometimes utilize lower air flow velocity to deliver the ions to the location where the ionization is required. Because of their lower recombination rates as compared with AC ionizers, DC ionizers are used in more contamination critical applications. In fact, for ISO Class 3 cleanroom operation, many DC ionizers perform effectively to the required Class 3 cleanliness level and are only limited by the turbulence and particulation issues of the fan and its motor.

DC ionizers are used in ISO Class 3 cleanroom applications. In this ultra-clean application, fan driven ionization cannot be used. In this case, the laminar air flow of the cleanroom is used for distribution of the ions throughout the room. DC ionizers used to provide ionization within a mini-environment or a process tool are called bars and provide a small aerodynamic cross-section to the laminar air flow. They contain one or more pairs of emitter points to produce the ionization.

Pulsed DC ionizers represent an alternate type of DC ionizer in an ISO Class 3 cleanroom, most commonly in the form of a ceiling-mounted ion emitter. An array of ceiling emitters is mounted to the ceiling of the cleanroom, typically at 1.2–1.8 m centers, and the laminar air flow from the HEPA filters distributes the ionization throughout the entire room. Normally the ionizers are operated in a mode called pulsed DC. This is very much like the steady state DC mode described above, but the two polarities are turned on one at a time with “on” times of 1–10 s (Figure 6.8). Pulsed DC provides even less recombination than steady state DC because the voltage is switched, maximizing the separation between the two polarities of ions. As a tradeoff, objects in the room swing

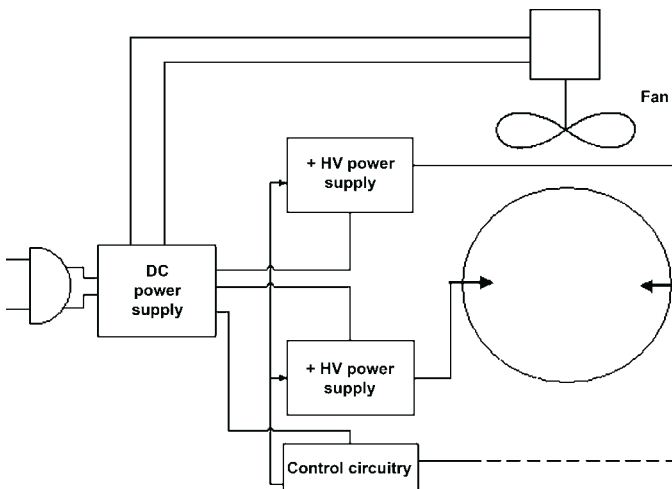


FIGURE 6.7 Schematic diagram of a DC ionizer.

through voltage levels of ± 50 to ± 150 V. This makes this method of ionization unacceptable for the most sensitive products such as MR heads, but ideal for applications where such sensitivity is not an issue and high levels of charge must be eliminated in a large work area.

6.6.2 Photoelectric Ionization

A photoelectric ionizer utilizes extreme UV to soft X-ray photons to create ions. When the energy of a photon exceeds the ionization potential of the medium it traverses, the photon can create an ion by collision with an atom. This is called the photoelectric effect.¹³ The most common method of photoelectric ionization involves the use of very soft X-rays. These are X-rays in the range of 5–10 keV, in contrast with medical or dental X-rays which are in the range of 125–175 keV. The lower energy makes them suitable for use as an ion source because the air is rather opaque to photons in this energy range. In contrast, medical X-rays must be chosen to be capable of traversing a human body with reasonable efficiency and, as such, have an attenuation length in air of hundreds of meters. Thus, medical X-rays are not suitable for creating ionization in a localized volume such as inside a process tool.

In contrast, most of the X-rays in the 5–10 keV energy range are absorbed in 1 meter of air as shown in Figure 6.9.¹⁴ A soft X-ray source has a 100% efficiency for creating an ion within about 1.5 m for each photon it emits. As such, it is capable of mediating very high levels of static charge. Although the device can be shielded by a few meters of air, it is regulated by governments in the countries it is used in. These regulations require licensing and also specify shielding requirements. While neither the licensing (in the USA by the FDA) nor the safety requirements are very difficult to manage, it does represent an extra step that is necessary for the use of this technology.

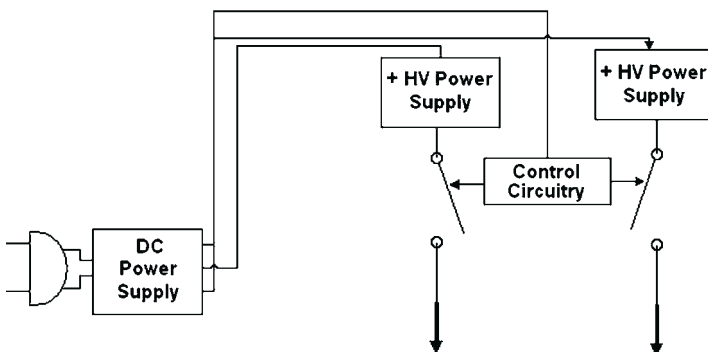


FIGURE 6.8 Schematic diagram of a pulsed DC ionizer.

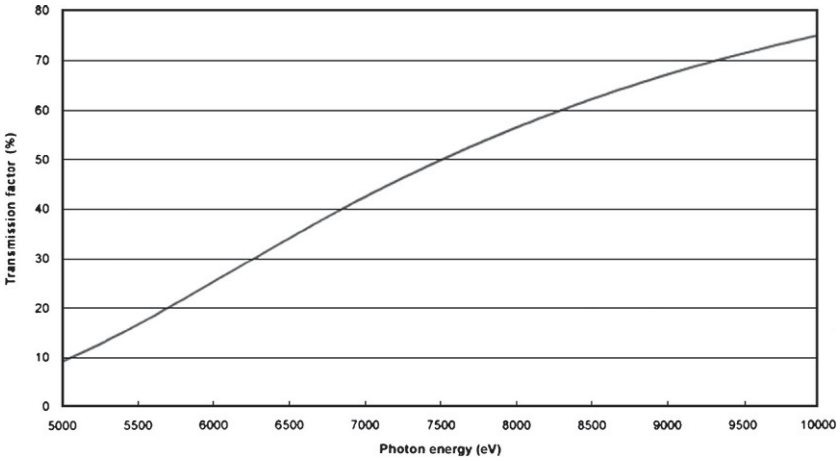


FIGURE 6.9 Transmission factor of photons through 50 cm of air at 1 atm.

6.6.3 Radioisotope Ionization

The alpha ionization method involves the use of ionizing radiation to make ions. While several forms of ionizing radiation sources are available, only α (alpha) radioactive sources are used for static control. The other radioactive sources have much longer penetration ranges and require shielding in impractical amounts. Most commonly, Po-210 is employed as the radioactive source because of its properties as an α emitter. It produces α particles with a range of only 3.8 cm in air and 0.02 mm in aluminum,¹⁵ so that virtually none of the α particles is emitted from the ionizing blower used at a work station, either through the air or through the chassis of the α source. Thus, α sources of this sort are regarded as harmless by government regulatory organizations in virtually every country. The other aspect of Po-210 that makes it an ideal choice is that it decays to Pb-206 which is a naturally occurring stable isotope. Unlike Po-210, many other sources decay into isotopes that may be active as well.

The reason that the range of α particles is so short is that they move much more slowly than any other natural radioactive source and thus, couple their energy much more efficiently to the air molecules, resulting in ionization. This makes an alpha type ionizer particularly efficient at creating ions; thus, it provides fast discharge times.

The greatest advantage of an alpha ionizer is that it does not employ HV to create the ions and thus, it does not emit electric fields, either AC or DC. Thus, α ionizers should have a balance voltage of zero, an important attribute in applications that involve very small and ESD sensitive products. This technology is crucial to the manufacture of disk drives with MR or GMR heads; as heads become smaller, it will be even more necessary to their manufacture.

The main disadvantages of the alpha ionizer systems are that personnel have a health concern about radioactive sources in their work environment. This disadvantage can be overcome with education of personnel. In addition, the device also requires some government paperwork for tracking and regulation of the radioactive material, and the sources must be replaced annually. This is not a financial issue but rather a logistical issue as the cost of replacement is comparable to the cost of maintenance required only of corona ionizers.

6.6.4 Measuring Ionizer Performance

Because the majority of ionizers use corona to generate ions, the ionizers involve HV. With HV come high electric fields. These fields are responsible for making the ions, but they also drive the voltage on objects around them to a non-zero voltage. While the voltage is most often much lower than the voltages caused by a static charge, it can be problematic nonetheless. For this reason, one of the parameters which must be measured to benchmark ionizer performance is the balance voltage. Loosely speaking, balance voltage is a measure of how accurately the positive and negative voltages of the ionizer can be set equal.

A CPM¹¹ is an instrument which is used to measure the performance of an ionizer. It includes a $15 \times 15 \text{ cm}^2$ ($6 \times 6 \text{ in}^2$) plate isolated from ground and with a capacitance of 20 pF as the voltage sensor. The balance voltage is the voltage to which the ionizer drives the plate. In addition, a CPM is used to measure the discharge time which is defined as the time to drive the plate from 1000 to 100 V.

6.7 AIR IONIZER APPLICATIONS

Static charge can cause problems anywhere in the cleanroom environment. To control these problems ionizers are used in a variety of configurations. Applications can involve cleanrooms which operate in the ISO Class 4 and 5 levels and are mostly involved with manual assembly. This is common, for example, for disk drive and flat panel assemblies. In contrast, disk drive media, flat panel cell and semiconductor front end cleanrooms are most often characterized by high levels of automation and typically require cleanliness levels at ISO Class 3 or better. These two scenarios require very different ionization solutions. Each of these situations will be discussed below.

The most common method used in ISO Class 3 ballroom-style cleanrooms is room air ionization. Ionizers are mounted at or near the filtered cleanroom ceiling and the laminar air flow is ionized as it passes the ionizer emitter points. The advantage of this type of system is that it controls static charge on any surface that is exposed to the room air flow. Corona ionizers are mounted in the ceiling to provide global coverage of the insulators moving through the room. These ionizers are either bars which hang from the ceiling, or they are ceiling emitters

mounted to the ceiling (and incorporating rods to stand off the emitter points from the ceiling). The distance that the emitter points are stood off from the ceiling is typically 30-100 cm. The purpose of this distance is to separate the sources of the ions from the grounded elements of the ceiling. If the ionizer emitters were flush mounted to the ceiling, all of the ions would be drawn to the grounded ceiling elements and none of the ions would be borne in the laminar air flow, providing electrostatic control for the insulators in the room. A typical room system will provide global coverage with discharge times (90% reduction in charge levels) in the range of 20–120 s. High ceilings and low air flow velocities represent the upper level and 2.4 m (8 feet) ceilings with 0.5 m/s (100 feet per minute air flow) provide the fastest discharge times.

Many cleanrooms today employ mini-environments for protection of the product from contamination. This configuration involves keeping silicon wafers in clean enclosures [standard mechanical interface (SMIFs) for wafers or reticles and front opening universal pods (FOUPs) for 300 mm wafers] as they are transferred from process tool to process tool. The individual tools are also fully enclosed to block contamination from the room. While this strategy blocks the contamination from contacting the product, it also blocks the ionization from contacting the products. Ions recombine as they pass through a HEPA filter, so room systems do not provide any electrostatic protection of objects within the process tools. To provide protection within the process tools, ion bars are mounted within the mini-environments as shown in [Figure 6.10](#). The bar, most typically a pulsed DC corona bar, should be mounted below the HEPA filter, if possible. To minimize offset voltages, the bar should be located at least 30 cm from the product. For applications requiring closer mounting, steady state DC bars should be used. If the direct HEPA location is infeasible (due to mounting particulars or due to a metal structure in the path of the air flow), a location should be chosen which is not immediately adjacent to a grounded metal structure, if possible. Care should be taken to select a location such that the air flow direction will carry the ions toward the object to be protected since the ions are not efficiently drawn against the wind electrostatically for more than a few centimeters ([Figure 6.11](#)).

The question often arises as to whether room ionization is necessary in a mini-environment cleanroom. In fact, from a contamination standpoint, most of the electrostatically driven particle deposition occurs in the mini-environments and in the carriers. The electric fields from objects outside of the carriers penetrate the carriers and add to the electric fields inside. As the carrier moves through the room or as charged objects in the room move past the carrier, the fields cause contamination within the carrier to move, representing a secondary contamination issue. Also, charged objects within the room, but external to the mini-environments, can drive a discharge from one object to another. While this does not affect the contamination control of the product



FIGURE 6.10 Ionizer applications in an ISO Class 3 cleanroom.

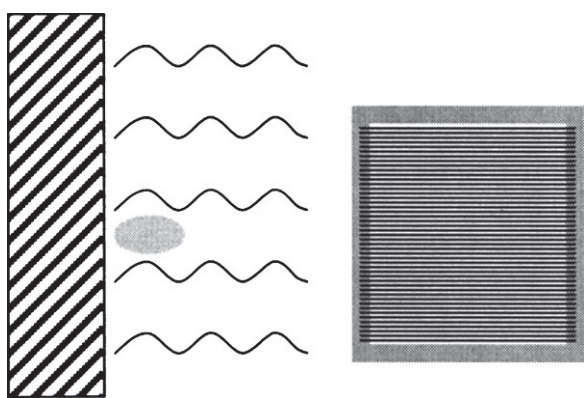


FIGURE 6.11 Utilizing ambient air flow to drive ions to the product to be protected.

directly, it does generate EMI in the room and can easily cause irrational performance by the robotics. For these two reasons, many people choose to install ionization both within the mini-environments and also in the cleanroom itself.

6.7.1 Discharge in Process Tools

Severe discharge requirements in process tools involving high levels of charge may employ the photoelectric ionizer, particularly in environments where it is possible to mount an ionization bar at a distance of 30 cm or greater from the product or in applications involving high levels of charge. Practically, this means either $>20,000$ V or highly capacitive objects. Cleaning processes tend to produce high charge levels. Also, processes that place flat products (like wafers or plates) on a flat surface, and then raise them up a significant distance when the processing is complete, are candidates for high charging.

In order to use the photoelectric ionizer, it is necessary to have an enclosure which serves as a complete shield against the photons. A rather thin shield is adequate (~ 5 mm of plastic or $750\text{ }\mu\text{m}$ of aluminum), but it is necessary to assure that there are no hairline cracks in the structure, such as doorways without an overlapping seal, as these can present a way for the photons to escape. In addition to the shielding requirement, there is a second requirement for a physical knife switch that shuts down the ionizer when the door is open. If these requirements are followed, it is very straightforward to get an FDA (United States Food and Drug Administration) X-ray cabinet license for this type of ionizer. While this small amount of paperwork is not required for a corona ionizer, there are applications of the sort mentioned above where this type of ionizer is the only solution which will provide protection for the process.

The other ionizer type which is used in tight confined spaces, or where static charge levels can be quite high, is the alpha ionizer. An alpha ionizer involves no HV so it has a balance voltage of zero, and it can be used with the most static sensitive products such as MR (magnetoresistive) or GMR (giant magnetoresistive) heads. An alpha ionizer requires air flow to deliver the ionization to the item to be protected, so it needs either a blower (discussed below) or it needs to be placed in a laminar flow field. However, it can be operated very close to the product with no significant effects. Since the range of Po-210 α particles in air is 3.8 cm, placing the ionizer within a distance of 5 cm of the charge source is perfectly acceptable. Again, at short distances an alpha ionizer is capable of dealing with extremely high charge levels. It requires minimal licensing, but, in practice, it does cause concern to manufacturing workers who do not understand it and it does require training by the RSD (radioactive source disposal) team on site.

6.7.2 Flow Benches and Work Surfaces

For clean laminar flow benches and other work surfaces in applications involving manual assembly such as in the disk drive industry, faster discharge times

(~5–20 s) are required to account for the rate of charging by objects being handled by operating personnel. In addition, the requirement for laminar air flow is not as great in this environment. Therefore, the ionization solution used for these applications involves forced air ionizers or ionizers with integral blowers. Most often, the ionizers are simply mounted on the table top and directed toward where the work is being done. While this can deal with most issues of static control, there are several practical issues that come into play in this environment. When objects are placed on a table top, they can block the airflow and, as such, render the ionizer useless as a static control device. Also, many operators find the air flow from the ionizer unpleasant due to the fact that the air makes them cold and causes their eyes to dry out. As a result, operators will occasionally turn off the ionizer when no one is watching, thus eliminating the protection that the ionizer provides.

The alternative to a table top ionizer is the use of an overhead ionizer. This is a better solution, although the cost is somewhat higher than a table top blower. An overhead blower ionizer is mounted above the work surface and blows air in the same direction as the laminar air flow (Figure 6.12). Since it is mounted

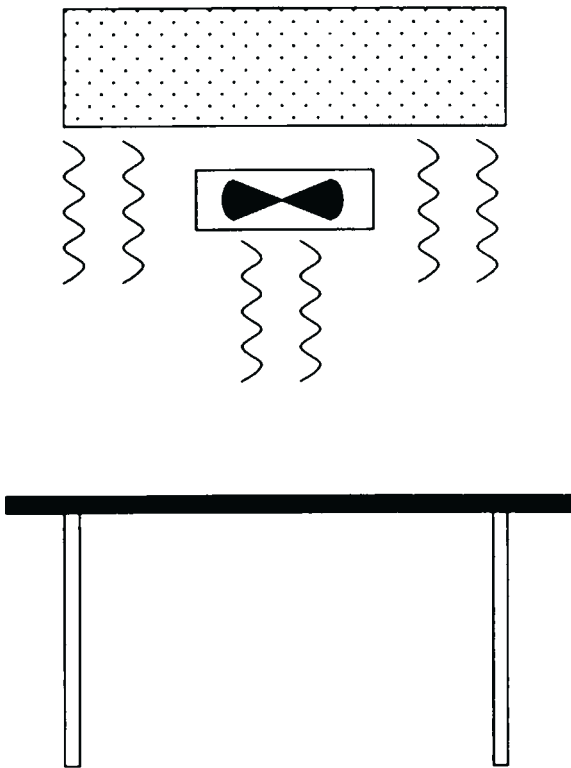


FIGURE 6.12 Directing the air flow downward to avoid blowing in the eyes of an operator.

above the work surface rather than on it, it is impossible to place objects on the work surface and block the air flow from reaching the product it is designed to protect. In addition, experience shows that this configuration eliminates the possibility of air flow into the operators' eyes by directing it downward rather than horizontally and thus eliminates the habit of turning off the ionizer.

The overhead ionizer typically achieves discharge times in the 5–15 s range, thus providing the ability to protect sensitive products that are being handled in a manual assembly area.

Alpha ionizers are commonly used in overhead blowers. They provide fast discharge times similar to corona ionizers but also feature zero-balance voltage. This type of ionizer is used almost exclusively in operations that involve handling of critically sensitive MR and GMR heads.

6.8 CONCLUSIONS

Reduction of product defects to levels acceptable in high quality, world-class cleanroom manufacturing requires continuing attention to static charge control. Uncontrolled, static charge is responsible for increasing the number of particle-related defects, damaging products and equipment by ESD, and interfering with factory automation. As factory profitability increasingly depends on both high yields and high product throughput, minimizing or eliminating static-related problems in cleanrooms has become a necessity.

Static charge control begins with the grounding of personnel, equipment, and any materials that come close to static sensitive products. Insulating materials should be removed, whenever possible, and replaced with static dissipative equivalent materials. When insulators must be used to satisfy other process requirements, ionizers are used to neutralize static charge on the insulators. It should be remembered that insulators are very often a part of the product itself and cannot be eliminated.

Air ionization is one of the few methods available for controlling static charge in high-quality cleanroom environments. In some cases it is the only method that can be used. Ionizers can reduce the number of contamination and ESD-related defects occurring in mini-environments and production equipment as well.

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